CHAPTER

03

Logic and Language

Correct reasoning can only be expressed **through language**, and many errors in logic come from careless or imprecise language. This chapter gives an overview of the philosophy of language, then studies **meaning** (cognitive and emotive meaning; vagueness and ambiguity; intension and extension), and finally **definitions**: their types, the techniques for making them, and the rules a good dictionary definition must follow.

LESSONS IN THIS CHAPTER

3.1 Philosophy of Language: An Overview

3.2 Logic and Meaning

3.3 Meaning, Types and Purposes of Definitions

3.4 Techniques of Definition

3.5 Criteria for Lexical Definitions

Freshman A+ · Logic & Critical Thinking (PHIL 1011)

CHAPTER 3 · LESSON 3.1

Philosophy of Language: An Overview

AFTER THIS LESSON YOU CAN

- Define philosophy of language and its questions
- Distinguish it from linguistics (Chomsky)
- Recall key figures in its history
- Name six theories of meaning

3.1.1 What is philosophy of language?

According to **semiotics** (the study of sign processes in communication and of how meaning is constructed and understood), language is the *use of symbols to draw attention to signified content*.

D PHILOSOPHY OF LANGUAGE

The reasoned inquiry into the origins of language, the nature of meaning, the usage and cognition of language, and the relationship between language and reality.

It asks questions like: *What is meaning? How does language refer to the real world? Is language learned or innate? How does the meaning of a sentence emerge out of its parts?*

Philosophy of language vs linguistics

Linguistics asks: *What makes English English? How do Spanish and French differ?* The great 20th-century linguist **Noam Chomsky** emphasized **grammar and syntax** (the rules that govern sentence structure).

★ CHOMSKY IN THREE POINTS

1. Humans are **born** with an innate **"universal grammar"**, a set of linguistic principles shared by all humans. Exposure to a particular language only **triggers** this knowledge, so language is primarily a matter of **nature**, not nurture.
2. He studies people's internal languages, **"I-languages"**, which are based on rules that generate grammars.
3. Other approaches, which he calls **"E-languages"**, explain a language as **usage** within a particular speech community.

Semantic holism: translation and interpretation raise problems that lead to the view that meaning belongs **not to a single word or sentence but to a whole language** (if at all).

3.1.2 A brief history

Period / thinker	Contribution
Plato	Names of things are determined by **nature**; each phoneme (the smallest unit that distinguishes meaning) represents basic ideas; convention plays only a small part

Period / thinker	Contribution
Aristotle	The meaning of a predicate comes from **abstracting the similarities** between individual things (later called **nominalism**); because those similarities are a real common form, he is also a **moderate realist**
Stoics	Five **parts of speech**: nouns, verbs, appellatives, conjunctions, articles. Their **lektón** (the sense of a term) led to the idea of the **proposition** (an assertion that can be true or false)
Medieval Scholastics	Saw **logic as a "science of language"**; studied vagueness, ambiguity, supposition, categorematic and syncategorematic terms
Renaissance	Interest in a **philosophical (universal) language**, spurred by the discovery of Chinese characters and Egyptian hieroglyphs
Late 19th–20th century	Language becomes **central**, especially in **analytic philosophy**

For **analytic philosophers**, philosophy of language has **four central problems**: the nature of **meaning**, language **use**, language **cognition**, and the relationship between **language, logic and reality**. **Continental philosophers** treat it as part of logic and other fields rather than as a separate topic.

3.1.3 What is meaning? Six theories

Meaning = *the content carried by the words or signs exchanged by people when communicating through language.* There are two essentially different types of linguistic meaning:

- **Conceptual meaning**: the definitions of words and the features of those definitions.
- **Associative meaning**: the speaker's individual mental understanding (connotative, collocative, social, affective, reflected or thematic).

Theory	Meaning is...	Associated with
Idea theories	**Mental contents** provoked by signs (semantic internalism)	British empiricists **Locke, Berkeley, Hume**
Truth-conditional theories	The **conditions under which an expression is true or false**	**Gottlob Frege**
Use theories	Related to **speech acts and particular utterances**, not expressions themselves	**Ludwig Wittgenstein**
Reference theories (semantic externalism)	The **things in the world** actually connected to signs	**Tyler Burge, Saul Kripke**
Verificationist theories	The **method of verifying or falsifying** a sentence	**Logical positivists** (early 20th century)
Pragmatist theories	Determined by the **consequences of its application**	Pragmatists

M "I TAUGHT UNCLE RICKY VERY PATIENTLY"

Idea (Locke…) · **T**ruth-conditions (Frege) · **U**se (Wittgenstein) · **R**eference (Kripke, Burge) · **V**erification (positivists) · **P**ragmatist (consequences).

Exam Practice

8 QUESTIONS · LESSON 3.1

1 Jigjiga University · Final

Noam Chomsky believed that language was:

A) Attributable primarily to nature
B) Attributable primarily to nurture
C) An equal mix of nature and nurture
D) Learned exclusively by watching parents

2 Salale University · Final 2014

_____ is the reasoned inquiry into the nature, origins and usage of language.

A) Linguistics
B) Philosophy of language
C) Moral philosophy
D) Semiotics only

3 Salale University · Final 2014

Fill in the blank: "_____ is the content carried by the words or signs exchanged by people when communicating through language."

4 Bahir Dar University · Model Exam

The theory that holds the meaning of an expression to be the conditions under which it may be true or false is associated with:

A) Kripke
B) Wittgenstein
C) Frege
D) Locke

5 Bahir Dar University · Model Exam

"Meaning is related to speech acts and particular utterances, not to the expressions themselves." This is the:

A) Verificationist theory
B) Use theory
C) Idea theory
D) Reference theory

6 Haramaya University · Model Exam

Which group distinguished five parts of speech, and whose concept of lektón gave rise to the idea of a proposition?

A) The Stoics
B) The Scholastics
C) The logical positivists
D) The Renaissance linguists

7 Haramaya University · Model Exam

The view that meaning cannot be attached to a single word or sentence but only to a whole language is called:

A) Nominalism
B) Verificationism
C) Universal grammar
D) Semantic holism

8 Jimma University · Model Exam

For continental philosophers, philosophy of language is usually treated as a separate topic rather than as part of logic.

True / False

Answers & Explanations

1. A

Attributable primarily to nature

Chomsky holds that humans are born with an innate universal grammar; exposure to a language only triggers it.

2. B

Philosophy of language

This is the module's definition of philosophy of language.

3. Meaning

The module's description of meaning (Lesson 1.3).

4. C

Frege

Truth-conditional theories go back to Gottlob Frege. Wittgenstein pioneered use theories; Locke idea theories; Kripke reference theories.

5. B

Use theory

Use theories were pioneered by Ludwig Wittgenstein.

6. A

The Stoics

The Stoics distinguished nouns, verbs, appellatives, conjunctions and articles. Their lektón led to the notion of a proposition that can be true or false.

7. D

Semantic holism

Translation and interpretation problems led to semantic holism.

8. **False**

The reverse is true: analytic philosophers treat it as a central topic, while continental philosophers deal with it as part of logic and other fields.

CHAPTER 3 · LESSON 3.2

Logic and Meaning

AFTER THIS LESSON YOU CAN

- Distinguish cognitive and emotive meaning
- Separate value claims from emotive language
- Tell vagueness from ambiguity and verbal from factual disputes
- Explain intension, extension and how they order terms

3.2.1 From words to arguments



words have meanings, conveyed through definitions

An argument is a group of **statements**; statements are **declarative sentences**; sentences are made up of **words**; and words have meanings conveyed through **definitions**. So **words are the most basic units** in any language and in every argument.

WATCH THE ORDER

"Arguments are composed of **words**, words are made up of **statements**..." is **false**. The order is argument → statements → sentences → words → meaning (Jigjiga Mid exam).

3.2.2 Cognitive and emotive meaning

Language has almost unlimited functions; **Wittgenstein** thought the number was virtually **unlimited** (asking questions, telling jokes, giving directions, greeting...). For logic, **two** functions matter most:

COGNITIVE MEANING

Terminology that **conveys information**.

"The death penalty, which is legal in thirty-six states, has been carried out most often in Georgia; however, since 1977 Texas holds the record for the greatest number of executions."

Words: *legal, thirty-six, most often, record.*

EMOTIVE MEANING

Terminology that **expresses or evokes feelings**.

"The death penalty is a cruel and inhuman form of punishment in which hapless prisoners are dragged from their cells and summarily slaughtered only to satiate the bloodlust of a vengeful public."

Words: *cruel, inhuman, hapless, dragged, slaughtered, bloodlust.*

The emotive example teaches **two points**:

1. Emotively charged statements usually have **both** cognitive and emotive meaning ("cruel" still means *tending to hurt others*). Logic is concerned **chiefly with cognitive meaning**, so we must separate the two.
2. Part of the cognitive meaning is a **value claim**: a claim that something is good, bad, right, wrong, better or worse. Here: *the death penalty is wrong*. **Value claims are often the most important part of the cognitive meaning** of emotive statements.

WHY PEOPLE USE EMOTIVE WORDS

Value claims normally **need evidence**. But when they are "dressed" in emotive language, the emotive **clothing hides the fact** that a value claim is being made and gives it **psychological momentum**. Listeners then "swallow the value claim whole" without evidence. Calling someone "crazy", "stupid" or "weird" is a lazy way of saying "what he does is wrong" without giving reasons.

Emotive terminology in arguments

In arguments, emotive language does the same job. It lets the arguer make **value claims without evidence**, and it gives the argument a **"steamroller" quality** that crushes counterarguments and paralyzes clear thinking. **The cure:** separate the value claims and other cognitive meanings from the emotive language and **rewrite them as distinct premises**.

REPHRASING AN EMOTIVE ARGUMENT (MODULE)

"Now that we know that the rocks on the moon are similar to those in our backyard and that tadpoles can exist in a weightless environment, and now that we have put the rest of the world in order, can we concentrate on the problems here at home?"

Neutral version: **P1** The space program has been confined to work on ordinary rocks and tadpoles. **P2** Rocks and tadpoles are less important than domestic hunger and unemployment. **P3** Our international efforts have restored order to every nation but our own. **P4** These efforts addressed less important problems. **C** The government should redirect these funds to domestic problems. Now we can see that P1, P3 and possibly P4 are **false**.

3.2.3 When cognitive meaning goes wrong: vagueness and ambiguity

	Vagueness	Ambiguity
Definition	**Borderline cases** where it is impossible to tell whether the expression applies	**More than one clearly distinct meaning** in a given context
Type of unclear	A **blur** of meaning: a **continuous range** of interpretations	A **mix-up** of otherwise clear meanings: **multiple discrete** interpretations
Word examples	love, happiness, peace, excessive, **fresh**, **rich**, **poor**, normal, conservative, polluted	**light**, **proper**, **critical**, stress, mad, inflate, chest, **bank**, sound, race
Statement example	"Today our job situation is more transparent."	"light pilsner": light in colour, in calories, or in taste?
Cured by	**Précising** definition	**Lexical** definition

! SAME WORD, BOTH PROBLEMS

"**Slow**" can be **ambiguous** (mentally slow or physically slow), and when it means *physically* slow it is still **vague** (how slow is slow?). Also: vagueness is not always a problem. Calling someone "tall" in conversation is fine; trouble comes only when the situation demands precision.

3.2.4 Verbal and factual disputes

Conflicts between people's arguments are called **disputes**.

★ VERBAL DISPUTE

Arises over the **meaning of language**. The conflict is **not genuine** and is resolved by **agreeing on how a word is to be understood**.

- **Vagueness:** Kassa says Mrs Zenebech *abuses* her children (she spanked one); Jemal says that is *discipline*. Where does discipline end and abuse begin?
- **Ambiguity:** Mullu says Dagim is *guilty* of cheating (he confessed); Worku says he is not guilty until convicted in court. **Moral** guilt vs **legal** guilt.

E FACTUAL DISPUTE

Arises over **a disagreement about facts**: the truth or falsity of claims.

Debebe: Aberash saw Fisseha steal a computer. Maru: Fisseha never stole anything; Aberash is lying to protect her boyfriend.

The issues are **factual**: did Aberash tell the truth, and did Fisseha steal?

★ PROCEDURE FOR ANY DISPUTE

Step 1: Is the dispute **verbal, factual, or a combination**? **Step 2:** If verbal, is the problem **vagueness or ambiguity**?

3.2.5 Terms: intension and extension

D TERM

Any word or arrangement of words that may serve as the subject of a statement. Terms include **proper names** (Abebe, the Ethiopian Parliament), **common names** (animal, house) and **descriptive phrases** (first Prime Minister of Ethiopia, author of *Oromay*).

Not terms: verbs, non-substantive adjectives, adverbs, prepositions, conjunctions and non-syntactic word strings (*dictatorial, runs quickly, above and beyond, moreover, craves*).

! USE VS MENTION

"**Wherever**" is not a term when it is **used** ("I will follow you wherever you go"). But when it is **mentioned** in quotation marks, it can be a subject: **"'Wherever' is an eight-letter word."** So any word can serve as a subject when it is mentioned.

Terms symbolize **two kinds of meaning**:

	Intension	Extension
Also called	**Connotation**, intensional meaning (≈ *sense*)	**Denotation**, extensional meaning (≈ *reference*)
Consists of	The **qualities or attributes** the term connotes	The **members of the class** the term denotes
"Cat"	furry, four-legged, moves in a certain way, meows	**All cats** in the universe: past, present, future

LOGIC ≠ GRAMMAR

In **grammar**, "connotation" means the subtle nuances of a word and "denotation" its direct meaning. **Logic uses the words differently**: connotation = attributes; denotation = class members.

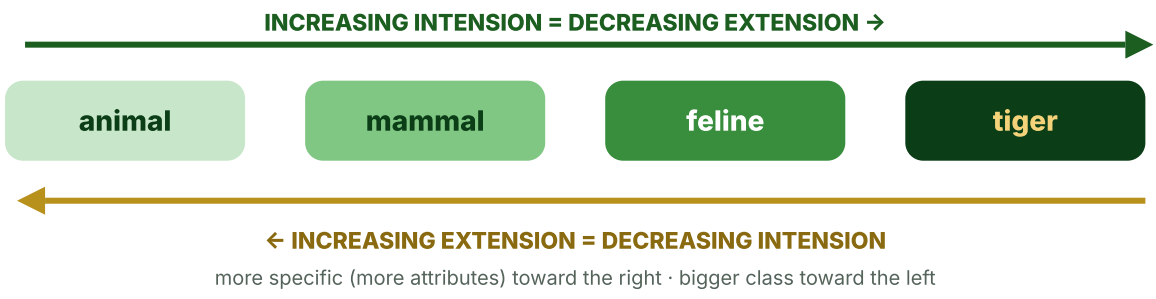
- **Conventional connotation:** to avoid subjective feelings (a cat-lover's "cuddly" vs a cat-hater's "disgusting"), logicians use the attributes a term commonly calls forth in **competent speakers**. So connotation stays **more or less the same from person to person and from time to time**.
- **Denotation can change** over time: "currently living cat" keeps changing; "cat" (all cats ever) is constant.
- **Empty extension:** "unicorn", "current king of France", "currently living dodo bird" denote the **empty (null) class**. They have empty **extension** but **not** empty **intension**.

INTENSION DETERMINES EXTENSION (NOT THE REVERSE)

The intension serves as the **criterion** for deciding what belongs to the extension. Because we know what "unicorn" means, we know nothing in the world fits it. **Extensions can suggest intensions but cannot determine them.**

Proper names seem to have no intension, yet they must have some, or we could not know whom they denote. Two solutions: (1) names are **shorthand for descriptions** ("Abebe" = the first-year civil engineering student...); (2) the intension is the **causal chain of events** from the naming ceremony through the community. Either way, **for all terms, intension determines extension**.

Ordering terms



more specific (more attributes) toward the right · bigger class toward the left

- **Increasing intension:** each term connotes **more attributes** (is **more specific**) than the one before: *animal, mammal, feline, tiger*.

- **Increasing extension:** each term denotes a **larger class**: *tiger, feline, mammal, animal*.
- Usually **increasing intension = decreasing extension** (and the reverse).

! THE EXCEPTIONS

1. **Empty extension:** "unicorn; unicorn with blue eyes; unicorn with blue eyes and a green horn..." → intension increases but extension cannot decrease (it is already empty).
2. **Same extension:** "living human being; living human being with a genetic code; ... with a brain..." → intension increases, but every term has **exactly the same** extension.

Exam Practice

20 QUESTIONS · LESSON 3.2

1 Jigjiga University · Mid 2016

Arguments are composed of words, words are made up of statements, statements have meaning, and meaning is conveyed through definitions.

True / False

2 Salale University · Final 2015

Value claims are often the most important part of the cognitive meaning of emotive statements.

True / False

3 Salale University · Final 2015

When the intension of a term with empty extension increases, its extension decreases.

True / False

4 Salale University · Final 2014

An expression is vague when it can be interpreted as having more than one distinct meaning in a given context.

True / False

5 Salale University · Final 2014

A term is ambiguous when there are borderline cases where it is impossible to tell whether it applies or not.

True / False

6 Oda Bultum University · Exam 2

Any word that cannot be a term when used can be a term when mentioned.

True / False

7 Oda Bultum University · Exam 2

Emotively charged terminology has both emotive meaning and cognitive meaning.

True / False

8 Oda Bultum University · Exam 2

Intensional meaning (conventional connotation) is more or less the same from time to time.

True / False

9 Jigjiga University · Final

Which of the following is INCORRECT?

A) The definiendum is the word that does the defining
B) Emotively charged statements have both cognitive and emotive meaning
C) Emotive terminology in arguments does basically the same job as in statements
D) The order of increasing extension is the same as that of increasing intension
E) A and B
F) C and D
G) A and D

10 Salale University · Final 2014

"The death penalty is a cruel and inhuman form of punishment in which hapless prisoners are dragged from their cells." Its main function is to:

A) Convey information
B) Express or evoke feelings
C) Give a précising definition
D) Report a statistic

11 Salale University · Mid 2015

The relationship between intension and extension is that:

A) Extension determines intension
B) Intension determines extension
C) They are unrelated
D) Every term has an empty extension

12 Salale University · Final 2015

Which is NOT a term with empty extension?

A) Current king of Ethiopia
B) Unicorn
C) Dinosaur
D) Current king of France

13 Oda Bultum University · Exam 4

Which of the following best illustrates "empty extension"?

A) Current Saudi Arabian king
B) Current Eritrean king
C) Current English king
D) All

14 Oda Bultum University · Exam 3

Which arrangement of terms is in the order of increasing intension?

A) Person – doctor of medicine – surgeon – brain surgeon – professional
B) Brain surgeon – surgeon – doctor of medicine – professional person – person
C) Person – professional person – doctor of medicine – surgeon – brain surgeon
D) None of the above

15 Oda Bultum University · Exam 4

A term is "vague" if it:

A) has multiple conflicting meanings
B) is intentionally misleading
C) lacks clear boundaries of application
D) is emotionally charged

16 Hurley · Exercise 2.1

Leslie: "Paul told us he would be visiting his parents in Knoxville this weekend, so he must not be at home." Diana: "I agree Paul is probably not at home, but you didn't hear him right. He said his parents live in Nashville." This dispute is:

A) Verbal (vagueness)
B) Verbal (ambiguity)
C) Factual
D) Not a dispute

17 Hurley · Exercise 2.1

Fred: "Today's professional athletes are overpaid. Many make millions a year." Shawn: "They are not overpaid at all. Look at the team owners; they make ten times as much." This dispute is:

A) Factual
B) Verbal: vagueness of "overpaid"
C) Verbal: ambiguity of "athlete"
D) No dispute

18 Hurley · Exercise 2.1

Frank: "That huge tree that fell last night must have made a tremendous crash." Shirley: "No. Sound is a perception, and perceptions depend on a perceiver. Nobody was here, so there was no crash." This dispute is:

A) Factual
B) Verbal: ambiguity of "sound"
C) Verbal: vagueness of "crash"
D) Not a dispute

19 Hurley · Exercise 2.2

Which series is in the order of increasing intension?

A) Sitka spruce, spruce, conifer, tree, plant
B) Plant, tree, conifer, spruce, Sitka spruce
C) Tree, plant, spruce, conifer, Sitka spruce
D) Conifer, plant, tree, spruce, Sitka spruce

20 Addis Ababa University · Model Exam

Kassa says Mrs Zenebech abuses her children because he saw her spank one; Jemal says spanking is discipline, which shows love. The dispute is:

A) Verbal, arising from ambiguity
B) Factual
C) Verbal, arising from vagueness
D) Not a dispute at all

Answers & Explanations

1. False

The order is reversed. Arguments are made of statements; statements are sentences; sentences are made of words; and words have meanings conveyed through definitions.

2. True

The module states this word for word.

3. False

An empty extension cannot get smaller. "Unicorn; unicorn with blue eyes..." increases in intension while the extension stays empty.

4. False

That is **ambiguity**. A vague expression has borderline cases.

5. False

That is **vagueness**. Ambiguity means multiple clearly distinct meanings.

6. True

"'Wherever' is an eight-letter word." A mentioned word can serve as the subject of a statement.

7. True

"Cruel" still means tending to hurt others, even while it expresses feeling.

8. True

Conventional connotation remains more or less the same from person to person and from time to time. It is **denotation** that may change with time.

9. G

A and D

A is wrong: the definiendum is the word **being defined** (the definiens does the defining). D is wrong: increasing extension runs in the **opposite** order to increasing intension. B and C are correct.

10. B

Express or evoke feelings

The module uses this exact sentence as its example of emotive meaning (cruel, inhuman, hapless, dragged).

11. B

Intension determines extension

The intension serves as the criterion for deciding what the extension consists of. The converse is not true.

12. C

Dinosaur

Like "cat", "dinosaur" denotes all its members, past and present, and dinosaurs really existed. (The original option A, "God", is left out because its answer depends on belief.)

13. B

Current Eritrean king

Eritrea has no king, so the term denotes nothing. Saudi Arabia and the United Kingdom both have kings.

14. C

Person – professional person – doctor of medicine – surgeon – brain surgeon

Each term is more specific than the one before. B is decreasing intension (increasing extension). A is out of order ("professional" at the end is general).

15. C

lacks clear boundaries of application

Vague terms have borderline cases. A describes ambiguity; D describes emotive meaning.

16. C

Factual

The disagreement is about a fact: did Paul say Knoxville or Nashville? (Hurley: factual.)

17. B

Verbal: vagueness of "overpaid"

Both agree about the salaries; they disagree about what counts as "overpaid", a vague word. (Hurley: verbal, vagueness.)

18. B

Verbal: ambiguity of "sound"

Does "sound" mean a subjective perception or an objective disturbance of the air? It is a mix-up of two clear meanings, so it is ambiguity. (Hurley: probably verbal, ambiguity.)

19. B

Plant, tree, conifer, spruce, Sitka spruce

Each term connotes more attributes than the one before (Hurley's answer). A is the order of increasing **extension**.

20. C

Verbal, arising from vagueness

Both agree that she spanked the child; the line between "discipline" and "abuse" is hazy, which is the module's own example of vagueness.

CHAPTER 3 · LESSON 3.3

Meaning, Types and Purposes of Definitions

AFTER THIS LESSON YOU CAN

- Define definition, definiendum and definiens
- Explain the five types of definition and their purposes
- Say which definitions can be true or false
- Classify a given definition

3.3.1 What is a definition?

D DEFINITION

A group of words that assigns a meaning to some word or group of words. Every definition has two parts:
- **Definiendum**: the word or group of words **being defined**.
- **Definiens**: the word or group of words **that does the defining**.

E SPOT THE PARTS

"**'Tiger'** means **a large, striped, ferocious feline indigenous to the jungles of India and Asia**." → definiendum = *tiger*; definiens = everything after "means".
Careful: the definiens is **not itself the meaning**. It is a group of words that **symbolizes the same meaning** as the definiendum.

M "-ENDUM" IS THE END PRODUCT BEING EXPLAINED

The definien**dum** = the word that **un**der**go**es defining. The defini**ens** = the words that do the defining.

3.3.2 The five types of definition (by purpose)

Type	Purpose	True or false?	Module example
Stipulative	Assigns a meaning to a word **for the first time**: a new word, or a new meaning for an old word; also **secret codes**	**Neither** (completely arbitrary)	**"Tigon"** = offspring of male tiger + female lion; **"liger"** = male lion + female tiger; "Tora, Tora, Tora"; "Operation Desert Storm"
Lexical	**Reports** the meaning a word **already has** in a language (dictionary definitions); **eliminates ambiguity**	**Yes** (true if it reports actual usage correctly)	Dictionary meanings of "nice", "light", "bank"

Type	Purpose	True or false?	Module example
Précising	**Reduces vagueness** so we can decide whether the word applies to a specific case	Not arbitrary; must be **appropriate** for its context	**"Poor"** = annual income below $4,000 and net worth below $20,000; "moment of death" = when the brain stops functioning
Theoretical	Assigns meaning by suggesting a **theory** about the things the term denotes, with deductive consequences and experiments	**Neither** (proposals); judged as **fruitful or not**	**"Heat"** = energy of the random motion of molecules; Leibniz's "substance" (monads); Mill's "good" (greatest happiness of the greatest number)
Persuasive	**Engenders a favourable or unfavourable attitude** toward what the definiendum denotes, using emotionally charged meaning	Can be true or false, but the main issue is **effectiveness** in persuading	"Abortion" = the ruthless murdering of innocent human beings; "Taxation" = the procedure bureaucrats use to rip off the people

"SLPTP": "STUDENTS LIKE PASSING TOUGH PAPERS"

Stipulative (new word) · **L**exical (dictionary) · **P**récising (cures vagueness) · **T**heoretical (a theory) · **P**ersuasive (an attitude).

Key details students confuse

STIPULATIVE VS PRÉCISING

Both assign a meaning, but a **stipulative** definition is **purely arbitrary** ("topar" would have worked as well as "tigon"), while a **précising** definition is **not arbitrary at all**: it must fit the context, as in law, science or medicine (is a bicycle a "vehicle"? when is the "moment of death"?).

LEXICAL VS PRÉCISING

Lexical definitions cure **ambiguity** (they list a word's distinct meanings). **Précising** definitions cure **vagueness** (they draw a sharp line for borderline cases).

- A stipulative definition gives **no new information** about the thing; it just names it. It can only be **more or less convenient** or appropriate.
- **Misuse of stipulative definitions:** in verbal disputes, a person uses a word in his own peculiar way and assumes everyone else uses it the same way.
- Whenever words from ordinary use enter **science, mathematics, medicine or law** ("force", "energy", "acid", "contract"), they must be given **précising** definitions.
- Theoretical definitions are not only scientific. Many philosophical terms (substance, form, cause, mind, God) have theoretical definitions.
- A persuasive definition is a **mix** of stipulative, lexical and possibly theoretical definitions, backed by the **motive to create an attitude**. It *masquerades* as an honest assignment of meaning, and it is common in **political speeches and editorials**.

Exam Practice

16 QUESTIONS · LESSON 3.3

1 Jigjiga University · Final

"'Smoker' means a rude and disgusting individual who callously emits noxious tobacco fumes into the air, threatening the health and comfort of everyone in the vicinity." This definition is:

A) Persuasive
B) Theoretical
C) Précising
D) Lexical

2 Jigjiga University · Final

A purely arbitrary assignment of meaning to a word is accomplished by a:

A) Lexical definition
B) Précising definition
C) Stipulative definition
D) Synonymous definition

3 Jigjiga University · Final

Bill Gates wants to help "the poor". Because "poor" is vague, a definition that specifies exactly who counts as poor is a:

A) Operational definition
B) Demonstrative definition
C) Précising definition
D) Theoretical definition

4 Jigjiga University · Final

You and a friend agree to say "One" to mean "True" so that you can communicate secretly in an exam. This definition is:

A) Lexical
B) Persuasive
C) Stipulative
D) Précising

5 Salale University · Final 2015

"'Magnetism' means a property of iron, cobalt and nickel that arises from the spin of the electrons in the unfilled inner shell of their atoms." This definition is:

A) Précising
B) Lexical
C) Persuasive
D) Theoretical

6 Salale University · Final 2015

"'Tiguar' means a breed between a male tiger and a female jaguar." This definition is:

A) Stipulative
B) Précising
C) Theoretical
D) Lexical

7 Salale University · Final 2015

"'Short' means a person who measures less than 1.5 meters." This definition is:

A) Persuasive
B) Theoretical
C) Stipulative
D) Précising

8 Salale University · Final 2015

A definition given to impose a certain attitude on readers or audiences is a:

A) Lexical definition
B) Theoretical definition
C) Persuasive definition
D) Stipulative definition

9 Salale University · Final 2014

One purpose of a lexical definition is to guard against the ambiguous use of a word.

True / False

10 Salale University · Final 2014

Stipulative definitions can be evaluated as true or false.

True / False

11 Oda Bultum University · Exam 2

"'Ramster' means an animal produced by crossbreeding a rat with a hamster." What type of definition is it?

A) Persuasive
B) Précising
C) Stipulative
D) Lexical

12 Oda Bultum University · Exam 2

"'Poor' means, for legal purposes, any person who has a daily income of less than $1." This is a:

A) Persuasive definition
B) Précising definition
C) Theoretical definition
D) Lexical definition

13 Hurley · Exercise 2.3

"'Blind' means, for federal income tax purposes, either the inability to see better than 20/200 in the better eye with glasses or having a field of vision of 20 degrees or less." This is a:

A) Lexical definition
B) Précising definition
C) Stipulative definition
D) Theoretical definition

14 Hurley · Exercise 2.3

"'Sound' means a compression wave, in air or some other elastic medium, having a frequency ranging (for humans) from 20 to 20,000 vibrations per second." This is a:

A) Persuasive definition
B) Stipulative definition
C) Theoretical definition
D) Lexical definition

15 Hurley · Exercise 2.3

"'Petrograb' means invading a country to steal its oil." This is a:

A) Lexical definition
B) Stipulative definition
C) Précising definition
D) Theoretical definition

16 Hurley · Exercise 2.3

"'Diffident' means lacking confidence in oneself; characterized by modest reserve." This is a:

A) Lexical definition
B) Persuasive definition
C) Précising definition
D) Stipulative definition

Answers & Explanations

1. A

Persuasive

Emotionally charged words (rude, disgusting, callously, noxious) aim to create an unfavourable attitude.

2. C

Stipulative definition

A stipulative definition is a completely arbitrary assignment of meaning for the first time, so it cannot be true or false. A précising definition is "not at all arbitrary".

3. C

Précising definition

The purpose of a précising definition is to reduce vagueness; the module's own example defines "poor" for financial assistance.

4. C

Stipulative

Stipulative definitions are used "to set up secret codes" (e.g., "Tora, Tora, Tora").

5. D

Theoretical

It characterizes the phenomenon through a scientific theory (electron spin), which suggests consequences and experiments, like the module's definition of "heat".

6. A

Stipulative

It coins a new word for a new thing, just like "tigon" and "liger".

7. D

Précising

"Short" is vague, and the definition draws a precise line. It is not stipulative, because "short" already has a meaning.

8. C

Persuasive definition

Its purpose is to engender a favourable or unfavourable attitude toward what the definiendum denotes.

9. True

Lexical definitions list a word's various meanings, which eliminates the ambiguity that would otherwise arise.

10. False

"There can be no such thing as a 'true' or 'false' stipulative definition." They are arbitrary.

11. C

Stipulative

It is a new word for a new thing.

12. B

Précising definition

It reduces the vagueness of "poor" for a legal context.

13. B

Précising definition

It sharpens a vague word for a legal purpose (Hurley: précising).

14. C

Theoretical definition

It conceives sound through a physical theory of waves (Hurley: theoretical).

15. B

Stipulative definition

It coins a new word (Hurley: stipulative). It also has a negative tone, but its main job is to introduce a new word.

16. A

Lexical definition

It reports the dictionary meaning of an existing word (Hurley: lexical).

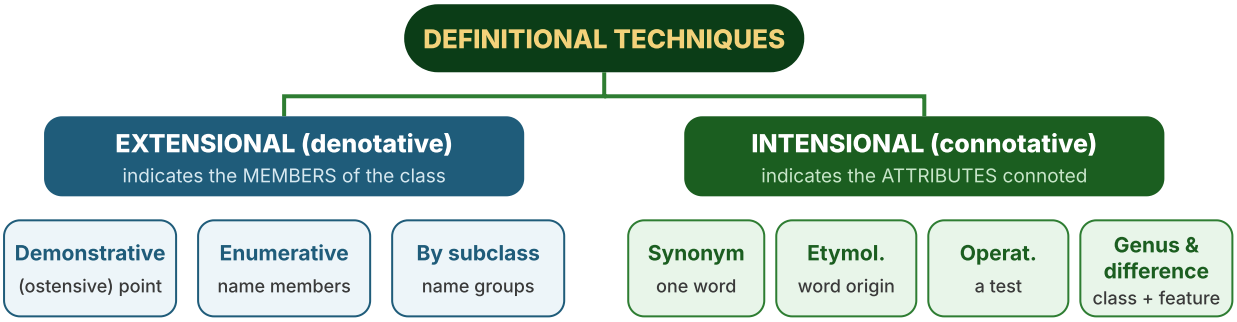
CHAPTER 3 · LESSON 3.4

Techniques of Definition

AFTER THIS LESSON YOU CAN

- Explain the three extensional techniques
- Explain the four intensional techniques
- Know which technique can produce which type of definition
- Explain why extensional definitions are deficient

Definitions can be produced by two families of techniques, matching the two kinds of meaning.



3 extensional + 4 intensional techniques = 7 techniques

3.4.1 Extensional (denotative) techniques

An **extensional definition** assigns a meaning by **indicating the members of the class** the definiendum denotes. Each may be **partial** (some members) or **complete** (all members).

Technique	How	Example
Demonstrative (ostensive)	**Pointing** to members	"'Chair' means **this** and **this** and **this**" (pointing); "'Washington Monument' means **that**"
Enumerative	**Naming members individually**	"'Actor' means a person such as Abebe Balicha, Samson Taddesse, or Mahder Assefa"
By subclass	**Naming subclasses** (kinds)	Partial: "'Tree' means an oak, pine, elm, spruce, maple, and the like". Complete: "'Fictional work' means a poem, a play, a novel, or a short story"

NOTES ON EXTENSIONAL DEFINITIONS

- **Demonstrative** definitions are the **most primitive** (you only need to understand pointing; useful with a foreigner) and the **most limited** (the object must be present: you cannot point to the sun at night). Their definiens is partly a **gesture**, which counts as a "word", as in sign languages.
- **Complete** enumerative and subclass definitions are more satisfying, but **few classes can be completely listed**.
- Extensional definitions are mainly used to produce **lexical and stipulative** definitions.
- They **cannot** serve as **précising** definitions: vagueness is a problem of **intension**, and specifying the extension would mean letting extension determine intension, which cannot be done.

ENUMERATIVE VS SUBCLASS

Ask: are the items **individual members** (Abebe, Germany, Ohio, French) → **enumerative**; or **kinds/groups** (oak, pine; poem, novel; doctor, lawyer) → **by subclass**. Hurley: "'Professional person' means a person such as a doctor, a lawyer, a professor..." is **by subclass**, because doctors are a *kind* of professional, not one person.

Why extensional definitions are deficient. Since intension determines extension (not the reverse), naming members can never guarantee the intended meaning:

- Pointing only at **wooden** chairs → someone may think "chair" means *wood*.
- Naming famous actors → someone may think "actor" means *famous person*.
- Listing kinds of trees → someone may think "tree" means *firmly planted in the ground* (which would include building pilings).

3.4.2 Intensional (connotative) techniques

An **intensional definition** assigns a meaning by **indicating the attributes** the word connotes.

Technique	How	Example	Limits
Synonymous	The definiens is **one word** with the same meaning	"'Physician' means doctor"; "'Observe' means see"	**Highly concise**, but many words have no exact synonym ("wisdom" ≠ knowledge, understanding, sense)
Etymological	Discloses the word's **ancestry** in its own and other languages	"'License' comes from Latin *licere*, to be permitted"; "'principle' from *principium*, beginning or source"	Gives the **root meaning**, and access to families of words (poly-gon → poly-gamy)
Operational	Specifies an **experimental procedure (test)** that decides whether the word applies	"A solution is an 'acid' **if and only if** litmus paper turns red when dipped into it"; "harder than" = scratches the other	Gives only **part** of the meaning; cannot define "love", "freedom", "dignity". *"pH below 7" is not operational: it prescribes no operation*

Technique	How	Example	Limits
Genus and difference	**Genus** (a larger class) + **specific difference** (the distinguishing attribute)	"'Ice' means **frozen water**"; "'Daughter' means **female offspring**"; "'Husband' means **married man**"	The **most generally applicable and adequate** intensional technique

★ **GENUS, SPECIES AND DIFFERENCE IN LOGIC**

In logic, **genus** = a relatively **larger class**; **species** = a relatively **smaller subclass** of the genus (genus animal → species mammal; genus feline → species tiger). The **specific difference** is the attribute that distinguishes the species. **Recipe:** pick a genus more general than the word, then narrow it with a difference until it means exactly the same thing.

Species (term)	Difference	Genus
"Ice"	frozen	water
"Daughter"	female	offspring
"Husband"	married	man

Which technique can produce which type of definition?

Technique	Can produce
Genus and difference	**All five** types (stipulative, lexical, précising, theoretical, persuasive); lexical definitions are typically of this kind
Operational	Stipulative, lexical, précising, persuasive (but not a *complete* lexical definition)
Synonymous	**Only lexical** (the word must already have a meaning, and the definiens adds no information)
Extensional (all three)	Mainly lexical and stipulative; **never précising**

M **"POINT, NAME, GROUP" | "SAME, ROOT, TEST, CLASS"**

Extensional: **Point** (demonstrative) · **Name** (enumerative) · **Group** (subclass).
Intensional: **Same** word (synonym) · **Root** (etymology) · **Test** (operational) · **Class** + difference (genus).

Exam Practice

17 QUESTIONS · LESSON 3.4

1 Jigjiga University · Final

"'Federal state' means countries such as Germany, Canada, India, Ethiopia, Switzerland, USA, Australia, etc." This definition is:

A) Enumerative
B) Ostensive
C) Definition by subclass
D) Synonymous

2 Jigjiga University · Final

Identify the ODD one.

A) Definition by genus and difference
B) Operational definition
C) Definition by subclass
D) Etymological definition

3 Jigjiga University · Final

"The word 'anatomy' is derived from the Greek word 'anatome', which means to cut up." This definition is:

A) Enumerative
B) Synonymous
C) Etymological
D) Operational

4 Jigjiga University · Final

"An 'electric current' flows in a circuit if and only if an ammeter connected in series shows a reading." This definition is:

A) Etymological
B) Demonstrative
C) Operational
D) Synonymous

5 Salale University · Final 2014

_____ is an intensional technique in which the definiens is a single word that connotes the same attributes as the definiendum.

A) Synonymous definition
B) Etymological definition
C) Operational definition
D) Enumerative definition

6 Oda Bultum University · Exam 2

In which definition is the definiens partly constituted by a gesture of pointing?

A) Enumerative definition
B) Ostensive definition
C) Demonstrative definition
D) Etymological definition
E) B and C

7 Oda Bultum University · Exam 2

A _____ definition is a highly concise way of assigning a meaning to a word.

A) Etymological
B) Synonymous
C) Ostensive
D) Operational

8 Oda Bultum University · Exam 3

"Philosophy is derived from two Greek words, 'philo' (love) and 'sophia' (wisdom)." What kind of definition is used?

A) Synonymous
B) Etymological
C) Operational
D) Genus and difference

9 Oda Bultum University · Exam 4

Which of the following is an example of an ostensive definition?

A) Defining "gold" as "a yellow metallic element with atomic number 79"
B) Defining "dog" as "Labrador, Poodle and Bulldog"
C) Defining "happiness" by pointing to a smiling person
D) Defining "justice" as "what is fair"

10 Salale University · Final 2014

Give an example of a definition by genus and difference and identify its parts.

11 Hurley · Exercise 2.4

"'Plant' means something such as a tree, a flower, a vine, or a cactus." This is a:

A) Enumerative definition
B) Definition by subclass
C) Genus and difference
D) Demonstrative definition

12 Hurley · Exercise 2.4

"'State' means something such as Ohio, Arkansas, Minnesota, and Tennessee." This is a:

A) Enumerative definition
B) Definition by subclass
C) Synonymous definition
D) Operational definition

13 Hurley · Exercise 2.4

"'Hot' means, for an electric iron, that your wetted finger sizzles when placed momentarily in contact with it." This is a:

A) Genus and difference
B) Operational definition
C) Précising lexical definition
D) Etymological definition

14 Hurley · Exercise 2.4

"'Hurricane' means a storm having constant winds of at least 74 miles per hour that originates at sea." This is a:

A) Definition by genus and difference
B) Operational definition
C) Enumerative definition
D) Synonymous definition

15 Hurley · Exercise 2.4

"'Professional person' means a person such as a doctor, a lawyer, a professor, or an architect." This is a:

A) Enumerative definition
B) Definition by subclass
C) Genus and difference
D) Demonstrative definition

16 Hurley · Exercise 2.4

"'Rapier' means sword." This is a:

A) Genus and difference
B) Synonymous definition
C) Etymological definition
D) Enumerative definition

17 Wollo University · Model Exam

Why can extensional definitions NOT serve as précising definitions?

A) Because vagueness is a problem of intension, and extension cannot determine intension
B) Because they are always too long
C) Because they are always persuasive
D) Because they cannot name any members

Answers & Explanations

1. **A**

Enumerative

It names individual members (particular countries), not kinds of federal state. The "etc." makes it a partial enumerative definition.

2. **C**

Definition by subclass

Definition by subclass is **extensional**. The other three are intensional techniques.

3. **C**

Etymological

It discloses the word's ancestry in another language.

4. C

Operational

It specifies a procedure (connect an ammeter and read it) that decides whether the term applies, like the litmus test for "acid".

5. A

Synonymous definition

"'Physician' means doctor." Enumerative is extensional, so it cannot be the answer.

6. E

B and C

"Ostensive" and "demonstrative" are two names for the same technique.

7. B

Synonymous

When a single equivalent word exists, a synonymous definition is the most concise.

8. B

Etymological

It gives the word's ancestry.

9. C

Defining "happiness" by pointing to a smiling person

Ostensive (demonstrative) definitions point. B is by subclass (kinds of dog), D is close to synonymous, and A is genus and difference.

10. **"Husband" means a married man.**

Species (the term defined) = husband; genus (the larger class) = man; difference (the distinguishing attribute) = married. Other examples: "ice" means frozen water; "daughter" means female offspring.

11. B

Definition by subclass

Trees, flowers, vines and cacti are **kinds** of plants, not individual plants (Hurley: by subclass).

12. A

Enumerative definition

Ohio and Arkansas are individual **members** of the class of states (Hurley: enumerative).

13. B

Operational definition

It prescribes a test to perform and observe (Hurley: operational).

14. A

Definition by genus and difference

Genus = storm; difference = constant winds of at least 74 mph, originating at sea. No test procedure is prescribed (Hurley: genus and difference).

15. B

Definition by subclass

Doctors, lawyers and professors are **subclasses** of professional persons, not named individuals (Hurley: by subclass).

16. B

Synonymous definition

The definiens is a single word (Hurley: synonymous).

17. A

Because vagueness is a problem of intension, and extension cannot determine intension

Making the intension precise by specifying the extension would mean extension determines intension, which is impossible.

CHAPTER 3 · LESSON 3.5

Criteria for Lexical Definitions



AFTER THIS LESSON YOU CAN

- State the eight rules for lexical definitions
- Identify which rule a bad definition breaks
- Recognize too broad and too narrow definitions
- Recognize circular, negative and affective definitions

Lexical definitions are the ones we meet most often; they are what most people mean by "the definition" of a word. So we need rules to **construct** and **evaluate** them.

Rule	A lexical definition should...	Bad example	Why it fails
1	**Conform to proper grammar**	"Vacation is **when** you don't have to go to work." · "Furious means **if** you're angry."	Grammatically incorrect. Better: "'Vacation' means a period during which activity is suspended from work or school."
2	**Convey the essential meaning**	"'Human' means a featherless biped."	Misses the essential attributes (capacity to reason and speak)
3	**Be neither too broad nor too narrow**	Too broad: "'Bird' = any warm-blooded animal having wings" (includes **bats**). Too narrow: "'Bird' = any warm-blooded feathered animal that can fly" (excludes **ostriches**)	The definiens includes too much / too little
4	**Avoid circularity**	"'Science' = the activity of scientists" + "'Scientist' = anyone who engages in science"; "'Silence' = the state of being silent"; "'Time' = whatever is measured by a clock"	Uses the word to define itself
5	**Not be negative when it can be affirmative**	"'Concord' means the absence of discord."	Better: "'Concord' means harmony". *Intrinsically negative words are fine: "bald" = lacking hair; "darkness" = absence of light*
6	**Avoid figurative, obscure, vague or ambiguous language**	Figurative: "'Camel' = a ship of the desert". Obscure: "'Bunny' = a mammalian of the family Leporidae...". Vague: "'Democracy' = government where the people are in control". Ambiguous: "'Triangle' = a figure of three straight lines in which all the angles are equal to 180°"	Paints a picture / needlessly technical / blurred / two readings

Rule	A lexical definition should...	Bad example	Why it fails
7	**Avoid affective terminology**	"'Communism' means that 'brilliant' invention of Karl Marx and other foolish political visionaries..."	Plays on emotions (sarcasm, facetious language)
8	**Indicate the context**	"'Strike' means (in baseball) a pitch at which a batter swings and misses" vs (in bowling) knocking down all pins	The meaning depends on the field

M **"GOOD ESSAYS NEVER CYCLE; POSITIVE, CLEAR, CALM, IN CONTEXT"**

Grammar · **E**ssential meaning · **N**either broad nor narrow · no **C**ycle (circularity) · **P**ositive · **C**lear (no figurative/obscure/vague/ambiguous) · **C**alm (no affective words) · **C**ontext.

! **DETAILS EXAMINERS TEST**

- Only **synonymous** definitions and definitions **by genus and difference** tend to be too broad or too narrow.
- **Operational** and **extensional** definitions alone cannot make a complete lexical definition. They should be **supplemented**, preferably by genus and difference (Rule 2).
- Technical language is not banned. **Needlessly** technical language is: "bunny" needs no zoology, but intrinsically technical words need technical definitions.
- Technically, the definiendum should be in **quotation marks or italics**.

Exam Practice

11 QUESTIONS · LESSON 3.5

1 Salale University · Final 2014

List at least three criteria for lexical definitions.

2 Hurley · Exercise 2.5

"A sculpture is a three-dimensional image made of marble." Which rule is violated?

A) Rule 3: too narrow
B) Rule 3: too broad
C) Rule 4: circular
D) Rule 7: affective terminology

3 Hurley · Exercise 2.5

"A cynic is a person who knows the price of everything and the value of nothing." (Oscar Wilde) Which rule is violated?

A) Rule 1: grammar
B) Rule 6: figurative language
C) Rule 5: negative
D) Rule 8: no context

4 Hurley · Exercise 2.5

"A theist is anyone who is not an atheist or an agnostic." Which rule is violated?

A) Rule 5: negative when it could be affirmative
B) Rule 4: circular
C) Rule 6: obscure
D) Rule 2: essential meaning

5 Hurley · Exercise 2.5

"'Truculent' is if you're cruel or fierce." Which rule is violated?

A) Rule 1: improper grammar
B) Rule 3: too broad
C) Rule 6: figurative
D) Rule 7: affective

6 Hurley · Exercise 2.5

"A cello is a stringed musical instrument played with a bow." Which rule is violated?

A) Rule 3: too narrow
B) Rule 3: too broad
C) Rule 5: negative
D) No rule is violated

7 Hurley · Exercise 2.5

"'Clock' means a manufactured device featuring two pointers that rotate past a set of numerals ranging from 1 to 12." Which rule(s) are violated?

A) Rule 2 (essential meaning) and too narrow
B) Rule 7 only
C) Rule 4 only
D) None

8 Hurley · Exercise 2.5

"'Pen' means an instrument used for writing on paper." This definition is:

A) Too broad only
B) Too narrow only
C) Both too broad and too narrow
D) Correct

9 Gondar University · Model Exam

"'Silence' means the state of being silent." Which criterion of lexical definition does this violate?

A) Indicate the context
B) Avoid circularity
C) Proper grammar
D) Avoid affective terminology

10 Gondar University · Model Exam

"'Bird' means any warm-blooded animal having wings." This definition is:

A) Affective
B) Too narrow, because it excludes ostriches
C) Circular
D) Too broad, because it includes bats

11 Arba Minch University · Model Exam

Which of the following negative definitions is ACCEPTABLE under Rule 5?

A) "Darkness" means the absence of light
B) "Concord" means the absence of discord
C) "Theist" means one who is not an atheist
D) "Health" means not being sick

Answers & Explanations

1. **Any three of the eight rules**

(1) Conform to proper grammar; (2) convey the essential meaning; (3) be neither too broad nor too narrow; (4) avoid circularity; (5) not be negative when it can be affirmative; (6) avoid figurative, obscure, vague or ambiguous language; (7) avoid affective terminology; (8) indicate the context.

2. **A**

Rule 3: too narrow

It excludes sculptures made of bronze, wood, plaster and so on (Hurley).

3. **B**

Rule 6: figurative language

It paints a witty picture instead of giving the essential meaning (Hurley).

4. **A**

Rule 5: negative when it could be affirmative

"Theist" can be defined affirmatively (one who believes in God), so a negative definition is inappropriate (Hurley).

5. **A**

Rule 1: improper grammar

"Is if..." is grammatically incorrect, like the module's "Furious means if you're angry" (Hurley: Rule 1).

6. **B**

Rule 3: too broad

The definiens also describes violins, violas and string basses (Hurley: too broad).

7. **A**

Rule 2 (essential meaning) and too narrow

It says nothing about the purpose of a clock (telling time) and excludes 24-hour and digital clocks (Hurley).

8. **C**

Both too broad and too narrow

It excludes pens used on glass, metal or plastic (too narrow) and includes pencils and crayons (too broad) (Hurley).

9. B

Avoid circularity

It defines the word by itself. It is the module's example of an intrinsically circular definition by genus and difference.

10. D

Too broad, because it includes bats

Bats are warm-blooded and winged but are not birds. "Warm-blooded feathered animal that can fly" would be too narrow (it excludes ostriches).

11. A

"Darkness" means the absence of light

Some words are intrinsically negative, and for them a negative definition is appropriate ("bald" = lacking hair; "darkness" = absence of light).

CHAPTER 3 · QUICK REVISION

Logic and Language — at a glance

Meaning and disputes

Concept	One-line meaning	Example
Cognitive meaning	Conveys information	"legal in thirty-six states"
Emotive meaning	Expresses or evokes feelings	"cruel, hapless, slaughtered"
Value claim	Something is good, bad, right or wrong	"The death penalty is wrong"
Vague	Borderline cases (a blur)	fresh, rich, poor, love
Ambiguous	Two or more distinct meanings (a mix-up)	light, bank, critical
Verbal dispute	About word meaning	"guilty" (moral vs legal)
Factual dispute	About facts	Did Fisseha steal it?
Intension	Attributes (connotation)	cat → furry, four-legged
Extension	Members (denotation)	all cats

Definitions: types and techniques together

Five TYPES (purpose)	Seven TECHNIQUES (method)
Stipulative: a new word, no truth value	**Extensional:** demonstrative (point), enumerative (name members), subclass (name kinds)
Lexical: reports usage, cures ambiguity, true or false	**Intensional:** synonymous (one word), etymological (origin), operational (test), genus and difference (class + feature)
Précising: cures vagueness, not arbitrary	Genus and difference → can make **all five** types
Theoretical: a theory, judged as fruitful	Synonymous → **lexical only**
Persuasive: creates an attitude	Extensional → **never précising**

! LAST-MINUTE CHECKLIST

1. Definiendum = the word defined; definiens = the words doing the defining.
2. Stipulative and theoretical definitions are neither true nor false.
3. Lexical → ambiguity; précising → vagueness.
4. Enumerative names **individuals**; subclass names **kinds**.
5. Increasing intension = decreasing extension, except for empty or equal extensions.
6. Connotation and denotation in logic ≠ in grammar.